THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS UNIT TEST SET B

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures; use graph paper where a graph is asked.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The volume of the largest sphere that can be carved from a cube of edge 7 cm is: 1

(a) 89.83 cm³ **(b)** 1437.33 cm³ **(c)** 179.67 cm³ **(d)** 343 cm³

2. The volume of a cylinder of radius 7 cm and height 10 cm is: 1

(a) 3080 cm³ **(b)** 1540 cm³ **(c)** 770 cm³ **(d)** 154 cm³

3. When a solid sphere of radius r is cut into two hemispheres, the increase in total surface area is: 1

(a) πr^2 **(b)** $3\pi r^2$ **(c)** $4\pi r^2$ **(d)** $2\pi r^2$

4. The total surface area of a solid hemisphere of radius r is: 1

(a) πr^2 **(b)** $3\pi r^2$ **(c)** $2\pi r^2$ **(d)** $4\pi r^2$

5. If $\Sigma f_i u_i = 20$, $\Sigma f_i = 50$, $a = 50$ and $h = 10$, the mean is: 1

(a) 46 **(b)** 52 **(c)** 50.4 **(d)** 54

6. For the frequencies 0–10: 4, 10–20: 7, 20–30: 12, 30–40: 9, 40–50: 3, the modal class is: 1

(a) 10–20 **(b)** 40–50 **(c)** 30–40 **(d)** 20–30

7. The most suitable measure to find the most popular shoe size sold in a shop is: 1

(a) mode **(b)** mean **(c)** median **(d)** range

8. For data with $n = 40$, Student A takes the median class as the class whose cumulative frequency first exceeds 20; Student B takes the class with the highest frequency. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

9. The probability of drawing a red king from a deck of 52 cards is: 1

(a) $\frac{1}{13}$ **(b)** $\frac{2}{13}$ **(c)** $\frac{1}{26}$ **(d)** $\frac{1}{52}$

10. The probability of a sure event is: 1

(a) $\frac{1}{2}$ **(b)** 0 **(c)** 1 **(d)** 100

11. Which of the following cannot be the probability of an event? 1

(a) 0.7 **(b)** $\frac{2}{3}$ **(c)** 115% **(d)** 0

12. A student states: "The probability of an event can be -0.2." The correct response is: 1

(a) Correct **(b)** Incorrect — $0 \le P(E) \le 1$
(c) Correct for impossible events **(d)** Cannot be determined

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

13. A solid is a cone (radius 6 cm, height 8 cm) mounted on a hemisphere of radius 6 cm. Find its total surface area. (Use $\pi = 3.14$) 2

14. A capsule is a cylinder with two hemispherical ends. Its total length is 21 mm and diameter 7 mm. Find its surface area. **2**

15. Find the median class and the modal class of: **2**

Class	0–20	20–40	40–60	60–80	80–100
Frequency	6	9	20	15	10

16. A piggy bank contains hundred 50 p coins, fifty ₹1 coins, twenty ₹2 coins and ten ₹5 coins. One coin falls out at random. Find the probability that it is **2**
(i) a 50 p coin
(ii) not a ₹5 coin.

17. A student says: "It will either rain tomorrow or it will not, so the probability of rain tomorrow is $\frac{1}{2}$." Justify or refute. **2**

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. An ice-cream cone is a cone of radius 3.5 cm and height 12 cm, filled and topped with a hemisphere of ice-cream of the same radius. Find the volume of ice-cream. **3**

19. Find the mean daily wage by the assumed mean method: **3**

Wages (₹)	100–120	120–140	140–160	160–180	180–200
Workers	12	14	8	6	10

20. Two customers Shyam and Ekta visit a shop in the same week (Tuesday to Saturday). Each is equally likely to visit on any day. Find the probability that they visit on **3**
(i) the same day
(ii) consecutive days
(iii) different days.

21. A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball is double that of a red ball, find the number of blue balls. **3**

22. The mode of the following data is 36. Find the missing frequency x. **3**

Class	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	8	10	x	16	12	6	7

OR

The mean of the following distribution is 62.8 and $\Sigma f = 50$. Find p and q. **3**

Class	0–20	20–40	40–60	60–80	80–100	100–120
Frequency	5	p	10	q	7	8

23. In a lottery there are 10 prizes and 25 blanks. Find the probability of **3**
(i) getting a prize
(ii) getting a blank.
(iii) If 5 more prizes are added, find the new probability of getting a prize.

OR

A box contains 100 cards numbered 2 to 101. One card is drawn at random. Find the probability that the number is **3**
(i) even
(ii) a perfect square
(iii) a prime less than 20.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. A bird-bath is in the shape of a cylinder with a hemispherical depression at one end. The height of the cylinder is 1.45 m and its radius is 30 cm. **5**

(i) Find the total surface area of the bird-bath (curved surface + bowl).

(ii) How many litres of water can the bowl hold?

OR

A shed is a cuboid 20 m × 10 m × 8 m with a half-cylinder roof of diameter 10 m running along its length. Find **5**

(i) the volume of the shed

(ii) the curved area of the roof.

25. Convert to a frequency table and find the median and mean: **5**

Marks less than	10	20	30	40	50
Students	3	8	17	20	23

OR

A student says the mode of the rainfall data below is 35 mm, since 30–40 has a large frequency. **5**

(i) Find the actual mode.

(ii) Find the mean.

(iii) Comment on the claim.

Rainfall (mm)	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Days	22	10	8	15	5	6	4

END OF QUESTION PAPER — Total: 50 Marks

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